



Lahore University of Management Sciences

AI623 – Deep Vision Language Models

Spring 2026

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Course URL (if any)	https://lms.lums.edu.pk/

Course Basics				
Credit Hours	3			
Lecture(s)	Nbr of Lec(s) Per Week	1	Duration	180 minutes
Recitation/Lab (per week)	Nbr of Lec(s) Per Week	-	Duration	-

Course Distribution	
Core	No
Elective	Yes
Open for Student Category	All meeting pre-reqs.
Close for Student Category	-

COURSE DESCRIPTION
This course offers a rigorous and research-aligned exploration of <i>Multimodal Deep Learning Models</i>, with a special focus on Diffusion Models for images and language generation tasks. We trace the evolution of multimodal deep models from CNNs to large language models (LLMs) and vision transformers (ViTs) to unified multimodal and generative architectures. We build on standard deep learning foundations, and focus on how modern AI systems integrate text, images, and other modalities within token-based Transformer frameworks. Students will study the mathematical and architectural principles behind decoder-only LLMs, cross-modal pretraining (e.g., CLIP), diffusion-based generation (e.g., Stable Diffusion, DALL-E 2, DLMs), and hybrid models that merge diffusion, flow, and autoregressive paradigms.

COURSE PREREQUISITE(S)
- MS AI Standing - or - CS331/EE315: Foundations of AI and Machine Learning, or equivalent course.
Note: All interested students will be asked to complete the zero-reading list and Assignment Zero individually. These materials will be released 10-15 days before the semester begins. The purpose of this material is to help level out differences in students' academic backgrounds. Submitting both the zero-reading list and Assignment Zero will be mandatory.

Grading Breakup and Policy
- Attendance: 5% - Programming Assignments (individual): 40% - Final Exam: 20% - Final (group) Project: 35%

Examination Details	
Midterm Exam	Yes/No: No Combine Separate: Duration: Preferred Date: Exam Specifications:
Final Exam	Yes/No: Yes Combine Separate: - Duration: 3 hours Exam Specifications: Open-book Open Notes



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Course Learning Outcomes

	At the end of the course, the students should be able to:
CLO1	Integrate and analyze modern architectures that unify language and vision, including Transformer-based and diffusion-based generative models, to understand their theoretical and practical foundations.
CLO2	Design and evaluate multimodal learning pipelines encompassing pretraining, cross-modal alignment, reasoning, and instruction tuning for diverse vision–language tasks.
CLO3	Critically assess emerging trends in hybrid, retrieval-augmented, and unified-token multimodal systems, applying advanced methods for model comparison, performance evaluation, and ethical alignment.

Relation to Program Learning Outcomes

CLOs	Related PLOs	Level of Learning	Teaching Methods	CLO Attainment checked in
CLO1	PLO1: Academic Education	Cog-2	Instruction, Homework	Assignment, Project
CLO2	PLO2: Knowledge for Solving Computing Problems	Cog-3	Instruction, Homework	Assignment, Project
CLO3	PLO2: Knowledge for Solving Computing Problems	Cog-4	Instruction, Homework	Assignment, Project

Textbook(s)/Supplementary Readings

Textbook: Class material

Reference material:

- Papers/books shared in the class

Week	Topics	Content
1-6	Foundations of Deep Generative Modeling and Diffusion Models	- Introduction to deep representation learning. 5 families of <i>Deep Models</i> based on gradient-signals taxonomy. - Why multi-modal and its core challenges? - Broader introduction to CLIP and why CLIP is a core multi-modal model but not a generative model. - From CLIP to Generative Modeling: VAE basics + ELBO Loss <u>Variational Perspective</u> - From VAEs to Diffusion: Basics of diffusion + stochastic process and SDEs - Types of diffusion models: DDPM, Improved DDPM variants. - Latent Diffusion <u>Score based Perspective</u>: Score concept in Diffusion, Denoising score matching, Comparison between DDPM and SM. <u>Score ODE framework</u>: Score SDE and its principles <u>Flow based perspective</u> and flow matching
7-8	Guidance and Controllable Generation	- Conditional generation mechanisms: classifier guidance and classifier-free guidance; guidance scale and trade-offs between fidelity and adherence to the prompt. - Controllability and conditioning: control via extra conditions (e.g., layout, edges, style, subject identity) and adapter-style modules (ControlNet/T2I-Adapter–type ideas). - Challenges and modern perspective.
9-10	CLIP, Unified Tokenization, and Multimodal Representation Spaces	- Models that bridge vision and language: CLIP (contrastive pretraining) image encoder, text encoder, contrastive InfoNCE objective. “Everything-as-tokens” paradigm: representing text, image in a unified sequence of discrete tokens. - VLMs with only text generation: what they can do and cannot do. - Text Tokenizers - Image tokenizers: VQ-VAE, VQ-GAN, Tokenizer Transformers. - Next token prediction LLMs and their training stages <u>Fusion strategies for VLMs</u>: early fusion, late fusion, cross-attention, adapters, shared latent spaces. - Representation geometry challenges: isotropy, collapse, and alignment in multimodal embeddings.



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11	Diffusion Models for Language	- Discrete diffusion for language. - Diffusion Language Models (noise-to-text generation paradigm, parallel denoising across positions, and comparison to autoregressive next-token prediction). - Recent advances for long-text and conditional generation. - Unified multimodal discrete diffusion. - Integration into joint vision–language spaces. - Comparison of autoregressive vs diffusion LLMs.
12-13	Reasoning in Multi-modal Context	- Reasoning multi-modal models - Reasoning in text vs reasoning in multimodal context - Reasoning from discrete token sequences to continuous latent spaces. - Multimodal CoT
14	Hybrid Generative Architectures	- Limitations of purely parametric multimodal models. - Memory transformers and key-value caches for multimodal contexts. <u>Recent hybrids</u>: Diffusion Transformers (DiT) and Flow-Matching Transformers (FMT). <u>Unified training objectives</u>: combining diffusion + autoregressive losses in shared latent space.

Academic Honesty

The principles of truth and honesty are recognized as fundamental to a community of teachers and students. This means that all academic work will be done by the student to whom it is assigned without unauthorized aid of any kind. Plagiarism, cheating and other forms of academic dishonesty are prohibited. Any instances of academic dishonesty in this course (intentional or unintentional) will be dealt with swiftly and severely. Potential penalties include receiving a failing grade on the assignment in question or in the course overall. For further information, students should make themselves familiar with the relevant section of the LUMS student handbook.

Harassment Policy

SSE, LUMS and particularly this class, is a harassment free zone. There is absolutely zero tolerance for any behavior that is intended, or has the expected result of making anyone uncomfortable and negatively impacts the class environment, or any individual's ability to work to the best of their potential.

In case a differently-abled student requires accommodations for fully participating in the course, students are advised to contact the instructor so that they can be facilitated accordingly.

If you think that you may be a victim of harassment, or if you have observed any harassment occurring in the purview of this class, please reach out and speak to me. If you are a victim, I strongly encourage you to reach out to the Office of Accessibility and Inclusion at oi@lums.edu.pk or the sexual harassment inquiry committee at harassment@lums.edu.pk for any queries, clarifications, or advice. You may choose to file an informal or a formal complaint to put an end of offending behavior. You can find more details regarding the LUMS sexual harassment policy [here](#).

To file a complaint, please write to harassment@lums.edu.pk.

SSE Council on Equity and Belonging

In addition to LUMS resources, SSE's **Council on Belonging and Equity** is committed to devising ways to provide a safe, inclusive and respectful learning environment for students, faculty and staff. To seek counsel related to any issues, please feel free to approach either a member of the council or email at cbe.sse@lums.edu.pk

Rights and Code of Conduct for Online Teaching

A misuse of online modes of communication is unacceptable. TAs and Faculty will seek consent before the recording of live online lectures or tutorials. Please ensure if you do not wish to be recorded during a session to inform the faculty member. Please also ensure that you prioritize formal means of communication (email, LMS) over informal means to communicate with course staff.